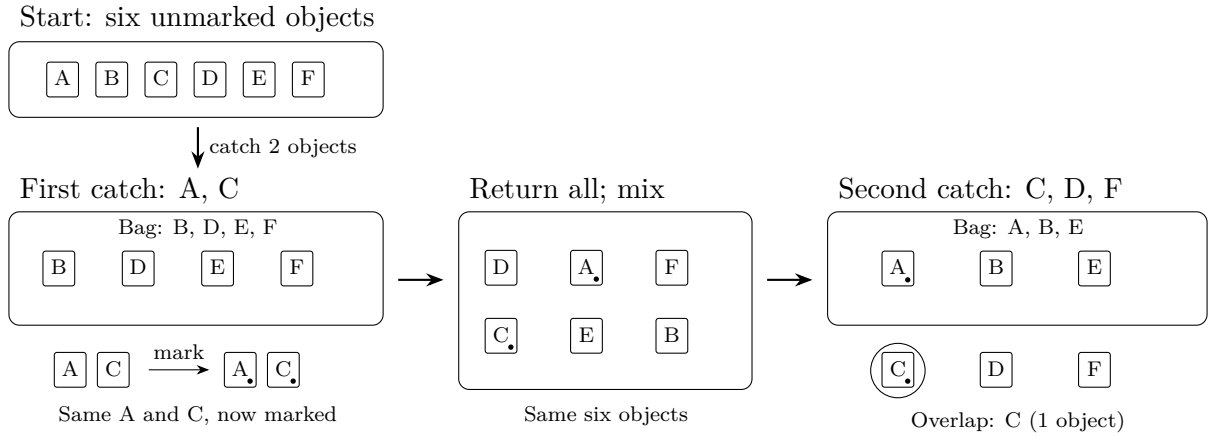
Use identical-size cards or counters with visible marker dots that cannot be felt. Keep the same objects throughout a series; add, remove or exchange none except for temporary catches. For sampling trials, draw without looking; every group of the chosen size should have the same chance, whether marked or unmarked.

Keep each catch out until it is counted. Mark the first catch, return *all* of it, and mix before the second catch. Marks stay on through both catches. An overlap is a marked object in the second catch. Before a new round, return everything and clear the marks. To show a record is possible, you may arrange catches by hand from the same bag; no object appears twice in one catch.



Problem 1: The keeper hides 8 to 20 objects. Catch 4, mark and return them, then catch 4 again. Play three rounds with the same objects. Choose a guess before the keeper reveals the total.

Round	Overlap	My guess
1		
2		
3		

Keeper's total: _____

Problem 2: Build the smallest bag that could give these catches. Build a larger bag that could give exactly the same catches. Which of these totals are possible: 6, 7, 8, 12, 20?

First catch	A	B	C	D	E
Second catch	B•	D•	E•	F	G

Problem 3: Now the two catches share no objects. Find the smallest possible bag. Is there a largest possible bag if you have as many cards as you want?

First catch	A	B	C	D	E
Second catch	F	G	H	I	J

Problem 4: These example rounds came from one unchanged bag. It holds either 8 or 16 objects. Every round has a fresh first catch of 4 and a second catch of 4. Which total would you guess from each round alone? Use all six rounds to choose one guess.

Round	Overlap	Guess: 8 or 16
1	2	
2	1	
3	3	
4	1	
5	2	
6	0	

My guess from all six rounds: _____

Problem 5: Make an 8-object bag and a 16-object bag. Arrange legal catches of 4 from each. Can either bag give both an overlap of 0 and an overlap of 3? Use your objects to settle it.

Problem 6: Use each bag below. The first catch has 5 objects and the second has 4. Find every overlap that can happen. Arrange catches to show your possibilities.

Bag total	Possible overlaps
6	
8	
12	

Problem 7: Choose a bag of 6, 7 or 8 objects. Make two catch records with first catch 5 and second catch 4: one that your bag could give and one that it could never give. Tell a partner the bag's total and the two records. Can they find the impossible record?